**AAE 52300: Introduction to Remote Sensing**
**3 credits**

## 1 Course Information/Description

Fundamentals of satellite and airborne remote sensing. Basic physical principles of electromagnetic wave propagation will be introduced. From this foundation, the phenomenology's enabling properties of the atmosphere; ocean and land surface to be measured at a distance will be developed. These principles will be applied to the design of instruments and satellite missions for Earth observation. Microwave instruments will be emphasized, although there will also be discussion of optical systems. Most of the material would also be applicable to remote sensing of other planets. Intended for students in Engineering or the Sciences.

## 2 Learning Resources and Reference Text

***Brightspace Learning Management System:***

Access the course videos, materials, and resources via Purdue's Brightspace learning management system.

***Required Textbook:***

Emery, Bill, and Adriano Camps, Introduction to Satellite Remote Sensing: Atmosphere, Ocean, Land and Cryosphere Applications, Elsevier, 2017, ISBN 978-0-12-809254-5, DOI:10.1016/C2015-0-04517-8

Available as e-book through Purdue Libraries.

***Optional Textbooks:***

Rees, Gareth, Physical Principles of Remote Sensing 3rd Edition, Cambridge University Press, 2013, ISBN 9781107004733

Chapman, Rick and Gasparovic, Richard , Remote Sensing Physics: An Introduction to Observing Earth from Space, Wiley/AGU, 2022, ISBN 978-1-119-66915-9

***Additional Readings:***

There will be additional readings for some of the sub-sections. All will be distributed through links to the specific module on the Brightspace page at the needed time.

***Software/web Resources:***

Students are expected to have access to and be familiar with the standard software used in engineering, including word processing, spreadsheets, presentation (e.g. Powerpoint) and computation (e.g MATLAB). Some example code and useful functions will be provided (mostly in MATLAB).

## 3 Assignments

### 3.1 Project

Each student will be required to complete a comprehensive literature review on a topic of their choice, selected in consultation with the instructor. If possible, distance students are advised to find a project relevant to their job. Students are also expected to make some individual contribution as part of their project. Examples of an individual contribution include: independent reproduction of a published result; comparison of different approaches to solving a problem; or applying a published method to a simple example problem. The project has four deliverables with due dates spaced at approximately equal intervals through the semester: (1) Abstract, (2) Literature review, (3) Draft report, (4) Final report. Feedback will be given for (1) (2) and (3), if they are submitted by the due date. Only the final report (4) will be evaluated to compute the final grade.

### 3.2 Homework

5 to 6 homework assignments will be given. Homework assignments will involve a substantial amount of computer programming and data processing. as a single PDF, giving a narrative describing the problem and the student's approach to solving it. All homework will be submitted through Brightspace.

## 4 Course Learning Objectives

1. Define essential variables of electromagnetic wave propagation and radiative transfer and apply them to describing the interaction of electromagnetic waves with the environment.

2. Explain the operation of remote sensing instruments, including active microwave (scatterometry, altimetry), passive microwave (radiometry), active optical (Lidar) and passive optical (multispectral, hyperspectral) instruments. Define the observables produced from them and show the application of these observables to estimate geophysical variables.

3. Describe the data processing chain from instrument function to final data products and apply this data processing approach to example problems.

4. Define the basic performance of remote sensing systems in terms of resolution and accuracy.

5. Design the orbit for a remote sensing satellite mission to meet basic coverage and revisit requirements.

6. Perform a literature review and write a scientific article.7. Demonstrate the use of a double-difference method in solving for the baseline between two receivers, using carrier phase observations.

8. Perform a literature review and write a scientific article.

## 5 Evaluation

Final grades should measure each student's level of understanding of the course material. Letter grades will be determined based upon a pre-defined scale (not relative to other students).

Homework: 50%

Final Project: 50%

The minimum requirement for "B": thorough review of current research literature on a topic selected by the student

Some individual contribution is required for the "A" grade. For example, applying a published algorithm to a simple problem, reproducing published results, or quantitatively comparing multiple approaches to a problem

The final numerical score will be directly converted to a letter grade according to the intervals in the following table:

A+: >=100 [i.e. extra credit is required]
A: [93, 100)
A-: [90, 93)
B+: [87, 90)
B: [83, 87)
B-: [80, 83)
C+: [77, 80)
C: [73, 77)
C-: [70, 73)
D+: [67, 70)
D: [63, 67)
D-: [60, 63)
F: <60